

Human Approval as a Controllable Simulation Mechanism

SocioVerse Challenge 2026 | Track C - Methodological and Systems Innovation

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Abstract

We implement a reproducible agent-based simulation of bounded approval capacity. Synthetic requests vary by risk, evidence completeness, a forbidden-action marker, arrival and deadline. We compare ungated execution, a static risk threshold and a scripted reviewer proxy. Twenty runs execute through the native SocioVerse2 engine (five prespecified seeds multiplied by four policy/capacity configurations). We report execution safety, valid-task completion, false denials and missed deadlines. **The results concern a deliberately simplified simulator, not real human reviewers or deployed AI agents.**

1. Research question and contribution

What is the measurable tradeoff between stopping unsafe synthetic actions and introducing review delay or mistaken rejection when review capacity is finite? The reusable contribution is a SocioVerse2-native Population / Environment / Decision / Collector implementation, a shared request generator and an auditable comparison protocol.

The comparison deliberately distinguishes model-free approval policy from model quality: the reviewer is a scripted mechanism, not a human subject or language model. Consequently this is a plumbing and methods test suitable for subsequent, more realistic experiments.

2. Hypotheses

H1: relative to no gate, explicit review of forbidden or risky requests decreases oracle-defined unsafe executions.

H2: a coarse static risk rule blocks some safe high-risk requests (false denials).

H3: reducing scripted reviewer capacity from eight to two requests per step increases deadline expirations and reduces safe-task completion.

No claims of operational benefit or causal effect on real organizations are made from these hypotheses or simulations.

Methods and experiment design

3. Population / Environment / Behavior (P/E/B)

Population. For each seed, 100 fixed synthetic requests with persistent identifiers, risk (high/low), evidence-completeness status, explicit forbidden marker, standard/urgent scheduling priority and a deadline. The identities are generated and contain no personal information.

Environment. Twelve arrival steps and a terminal-resolution horizon of 18 steps. Every comparison arm sees the same requests and initial state for each seed. The capacity parameter limits high-risk review decisions per step; late pending work expires.

Behavior. The no-gate arm executes immediately. Static policy denies all high-risk actions and executes lower-risk requests. The scripted-review arm immediately denies explicitly forbidden actions, allows low-risk requests and puts remaining high-risk requests into a bounded review queue. The proxy checks the synthetic evidence flag, not a human judgment.

4. Oracle and metrics

A request is oracle-safe when it is not explicitly forbidden and does not combine high risk with missing evidence. This definition is created by the simulation designer and is a key source of optimistic bias. Unsafe-action rate uses executed actions as its denominator; safe completion and false-denial rates use all oracle-safe requests as denominator.

Additional metrics include review expirations, latency in discrete simulated steps, the number of terminal outcomes, and breakdowns for the artificial urgency groups. They are not demographic fairness findings.

5. Prespecified protocol

Paired seeds: **11, 29, 47, 83 and 101**. A request set is generated once per seed and reused by every policy arm. Four variants: ungated, static, review capacity 8 and review capacity 2. All simulations run on SocioVerse2's native Python engine and persist DuckDB trajectories. An independent standard-library runner writes CSV/JSON metrics and integrity hashes.

All decisions use scripted logic, with **zero LLM calls** and an estimated model-API cost of **US \$0**. There are no external human participants, private records or unlicensed data.

Results: descriptive synthetic outcomes

6. Aggregate rates across the five seeds

Policy / condition	Unsafe rate	Safe tasks	False denials	Avg. expired
No gate	19.0%	100.0%	0.0%	0.0
Static rule	9.3%	68.6%	31.4%	0.0
Capacity review: 8	0.0%	100.0%	0.0%	0.0
Capacity review: 2	0.0%	94.0%	6.0%	6.0

Each row averages five deterministic synthetic replications of 100 requests each. The rates are descriptive; the five random seeds are not a representative sample of real workplaces.

7. Interpretation

Ungated execution processes all incoming requests but allows oracle-defined unsafe requests. The static rule reduces many unsafe executions but rejects some oracle-safe high-risk requests. The review proxy performs better by construction because it reads the very flags used to define the oracle. At limited capacity, this proxy cannot finish every safe request by its deadline.

The difference between capacity 8 and capacity 2 measures queue mechanics, not human cognition. Review performance is therefore a **best-case scripted baseline** under the chosen oracle, not evidence that human reviewers would reach the same rates.

8. Reproducibility and QA evidence

The source repository contains Python execution code, native provider adapters, P/E/B configuration files, the exact seeds, per-request trajectories, SHA-256 file digests and a 20-run native summary. Eight unit tests check deterministic replay, identical paired workloads, terminal uniqueness, policy invariants and audit boundaries. The native engine passed all 20 configured runs.

Public repository: github.com/ondmindmanagement-hub/unfire-socioverse-approval-gates

Limitations, validity and next work

9. Threats to validity

Construct validity. The oracle, requests and reviewer proxy share a deterministic synthetic rule. Apparent safety is partly tautological. The values are not calibrated to organization-specific policies.

External validity. No observed human approval times, organizational incentives, realistic malicious actors, LLM errors, asymmetric information or contested decisions are represented.

Statistical validity. Five seeds explore generator variation but do not produce population-level inference. No claim of statistical significance or effect sizes for real populations is warranted.

Operational validity. The demo omits reviewer authentication, override protocols, signed audit logs, privacy protection in actual user data, model latency, token expense and real-world integration risk.

10. Ethics and data use

All records are synthetic, without health data, financial account data, client events, identifying personal information or material copied from third-party users. No subjects are recruited. Priority labels are operational variables, not proxies for protected human traits.

11. Next experiments

First, add noisy reviewer classifications and intentional misclassification to avoid a perfect oracle. Second, vary action prevalence, review capacity, time limits and evidence availability across a broader grid. Third, include a human-controlled approval UI only after an ethical and methodological review. Fourth, compare real SocioVerse2 agent policies or LLM-controlled behavior with the scripted baseline while recording tokens, latency and cost.

12. Execution and submission status

At the SocioVerse2 repository root (Python 3.11+): `python -m studies.unfire_approval_gates.run_native`. Separately, `python -m unittest discover -s studies/unfire_approval_gates/tests`. The preliminary two-page proposal is ready but the competition-platform upload has not been confirmed. The final competition package additionally requires a short actual-run video and official submission before 31 October 2026 (UTC+8).

References and basis

SocioVerse2 open-source platform and Challenge 2026 official rules, socioverse.fudan-disc.com/challenge/; SocioVerse2 source repository (sii-research/SocioVerse2, Apache-2.0). Grimm et al. (2020), ODD protocol for describing agent-based and other simulation models, JASSS 23(2).

This document is a provisional research artifact, not a certified safety assessment or proof of competition acceptance.